

Calculus Homework – Differential calculus.

Exercise 1. Calculate the first and second order derivatives of the following functions!

$$(1.1) f(x) := (x + 2)^2 \quad (1.2) f(x) := x^3 + 5^x \cos(x) \quad (1.3) f(x) := (x^2 + 1)^{\cos(x)}$$

$$(1.4) f(x) := \ln(x^3 + 3x^2 + 3x + 1) \quad (1.5) f(x) := \tan(x^2) \quad (1.6) f(x) := \frac{x^2}{\log_5(x)}$$

$$(1.7) f(x) := \cos(\sin(x) + 1) \quad (1.8) f(x) := \sqrt{x} \ln(x) \quad (1.9) f(x) := \frac{x^2 + 4x + 4}{(x + 2)^3}$$

$$(1.10) f(x) := (x^3 + 1)^{x^3+1} \quad (1.11) f(x) := 3^{x+1} \sinh(x^4 + 2x) \quad (1.12) f(x) := x^x$$

$$(1.13) f(x) := \arctan(x) \quad (1.14) f(x) := \cosh(3x + 1) \quad (1.15) f(x) := \frac{4^x(x + 2)^3}{\ln^x(x)}$$

Exercise 2. Let $I \subseteq \mathbb{R}$ be a non-empty open interval and $f : I \rightarrow \mathbb{R}$ be differentiable at the point $p \in I$. Show that f is continuous at the point p ! Give an example for a function which is continuous everywhere but is not differentiable at some point of its domain!

Exercise 3. Let $k \in \mathbb{Q}$ be arbitrarily fixed and $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) := x^k$. Show that $f'(x) = kx^{k-1}$.

Exercise 4. Calculate the n th order derivative of the following functions!

$$(4.1) f(x) := 2^x \quad (4.2) f(x) := \cos(x) \quad (4.3) f(x) := \frac{1}{2x + 1} \quad (4.4) f(x) := \sin(3x)$$